

Knowledge Editing via Self-Study Data Synthesis

Abschlussbericht Projektarbeit

Iraj Masoudian, Heinrich-Heine-Universität Düsseldorf

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1 Methodology

The goal of this project is to inject factual edits into a language model by training on synthetic conversations generated with the Self-Study procedure of the Cartridges framework [1], and to compare this approach with dedicated knowledge editing methods such as ROME [2], MEMIT [3], and AnyEdit [4].

1.1 Benchmark

All experiments use the CounterFact split of the AKEW benchmark [5] with 975 counterfactual edits. Each entry provides the subject, the old (true) answer, the new (edited) answer, one base edit question, two paraphrases, and unstructured sentence representations of the fact.

1.2 Data Synthesis with Self-Study

Self-Study is a loop between two agents: a user bot asks questions about a given context, in our setting a single CounterFact edit, and an assistant bot answers them. Both are instances of the teacher model (**Qwen3-4b**, served with the Tokasaurus inference server). Every conversation starts from a seed prompt, an instruction from which the user bot derives its first message.

The data sources shipped with Cartridges handle plain text corpora. A new resource for knowledge editing was therefore implemented: it loads the CounterFact entries, extracts the subject, the relation, and the old and new answer of each edit, derives the relation phrase automatically from the CounterFact prompt template, and passes the unstructured fact sentence to the synthesizer as the conversation context.

The seed prompts that ship with the Cartridges framework target general corpora and cover generic tasks such as structuring, summarization, question answering, use cases, and creative writing; they are not suited for targeted fact edits. As part of this project, new seed prompt types were therefore designed specifically for knowledge editing. Each is instantiated with the subject, the relation, and the old or new answer of the current edit:

- *Negation*: the user asks whether the old fact still holds or asserts it with varying confidence; the assistant denies it and states the new fact.
- *Correction*: the user insists on the old fact and is corrected.
- *Question*: the user asks directly for the edited relation, in varied phrasings and registers.
- *Strict blocking*: the user probes the former connection between the subject and the old answer.
- *Creative*: an adapted variant of the generic creative type that explores consequences of the edit.

The pipeline itself was modified in several points. Each seed type now injects its own system prompt, which shapes the assistant’s behavior for that conversational pattern and instructs it to present the edit as its own knowledge; originally all seed prompts shared one generic system prompt. The seed prompt is also no longer passed to the user bot verbatim as its first message: it only describes an intent, which the user bot rewrites into a natural message in the voice of a randomly assigned persona, sampled at a high temperature of 0.95 for lexical diversity. The assistant answers at a low temperature of 0.4; assistant thinking is enabled with a fixed probability. With this setup, 4,096 samples per seed type were generated, 16,384 in total, so every edit appears about 17 times (4 to 5 samples per seed type and entry). In addition, single entry datasets were created to train per edit cartridges that can be concatenated.

1.3 Training

Two variants were trained on the synthesized data. The cartridge setup consists of one large cartridge containing all 975 edits and several small cartridges containing one edit each, which can be composed at inference time. Cartridges are initialized from the KV cache of a prompt; a new initializer was written that builds this prompt directly from a CounterFact entry as a set of behavior rules (state the new answer, correct mentions of the old one, do not reconnect the subject with the old answer) instead of generic corpus text. As a comparison, a LoRA adapter [6] was trained on the same data (configuration in Table 1).

Model & LoRA		Training	
Base model	Qwen3-4b	Batch size	4
LoRA r	16	Grad. accumulation	8
LoRA α	32	Effective batch size	32
LoRA dropout	0.05	Epochs	5
Max length	2048	Learning rate	$2 \cdot 10^{-4}$
Precision	BF16	Warmup steps	100
Train samples	14,745	Eval samples	1,639

Table 1: LoRA training configuration of the interim results.

1.4 Evaluation

Evaluation follows two tracks. First, an LLM judge scores model responses from 0 to 5 along four standard dimensions: efficacy (original question, is the edited fact returned), generalization (paraphrased question), locality (does the edit leak into unrelated facts), and portability (can the model reason on top of the edit). Second, BERTScore and ROUGE-L are computed against the reference answers, separately for original and paraphrased questions, matching the AKEW evaluation protocol used by AnyEdit [4].

2 Interim Results

Table 2 shows both variants next to the published AKEW CounterFact results for Qwen2.5-7B-Instruct.

Method	Original		Paraphrased	
	BERTScore	ROUGE-L	BERTScore	ROUGE-L
Pre-edited	65.50 ± 0.34	18.24 ± 0.60	44.74 ± 0.41	17.29 ± 0.51
FT-L	46.66 ± 0.48	14.63 ± 0.58	32.34 ± 0.50	12.31 ± 0.62
MEND	69.54 ± 0.54	25.47 ± 0.49	52.86 ± 0.40	22.81 ± 0.54
ROME [2]	75.89 ± 0.38	36.42 ± 0.45	55.67 ± 0.47	25.79 ± 0.59
MEMIT [3]	77.19 ± 0.32	38.95 ± 0.48	56.04 ± 0.40	25.73 ± 0.57
AlphaEdit	80.66 ± 0.25	45.55 ± 0.37	56.99 ± 0.49	27.69 ± 0.59
AnyEdit [4]	98.08 ± 0.15	95.09 ± 0.19	65.40 ± 0.38	43.49 ± 0.47
UnKE	97.34 ± 0.13	90.44 ± 0.16	59.29 ± 0.48	29.27 ± 0.61
AnyEdit*	99.63 ± 0.09	98.99 ± 0.10	60.78 ± 0.39	32.95 ± 0.59
LoRA (ours)	87.90	29.71	86.35	22.45
Cartridge (ours)	90.15	44.92	87.56	27.31

Table 2: AKEW CounterFact results (BERTScore / ROUGE-L, original and paraphrased questions). Baseline rows are the numbers published in [4] for Qwen2.5-7B-Instruct; the last two rows are interim measurements from this project.

Both variants reach higher BERTScore values than the locate and edit baselines (MEND, ROME, MEMIT, AlphaEdit), and the cartridge comes close to AlphaEdit on the ROUGE-L of the original questions. The cartridge performs better than the LoRA adapter trained on the same data in all four columns. The difference between original and paraphrased questions is small for both variants (cartridge: 90.15 vs. 87.56 BERTScore), while the baselines lose considerably more under paraphrasing. AnyEdit remains well ahead on the original questions. Two caveats apply to these numbers: the LLM judge frequently produced unparseable JSON output, so judge scores are not reported here, and both variants finetune Qwen3-4b while the baseline rows are published for Qwen2.5-7B-Instruct, so the comparison is only indicative.

3 Bugfixes and Optimizations

After the interim results, the pipeline was revised in the following points.

- **Judge output.** The judge now runs on a vLLM server with grammar constrained decoding (xgrammar), which guarantees valid JSON. The parsing failures that previously invalidated a large share of the scores are eliminated, and every evaluation run is gated by a calibration test with known answers.
- **Training data contamination.** An audit of the synthesized dataset showed two defects: raw `<think>` blocks stored as training targets (about 20 percent of the samples) and answers that reference their source (“according to the provided information”, about 25 percent). Thinking is now disabled during synthesis, the negation and question system prompts were rewritten so that the assistant states facts directly without citing a source, and a filter removes remaining violations after synthesis.

- **Teacher fidelity.** A second audit showed that in some answers the teacher keeps the old fact, often by presenting old and new fact as compatible. Every sample is therefore checked by the LLM judge, and answers that still assert the old fact are removed (13.3 percent of the regenerated dataset).
- **Edit coverage.** Edits are now drawn round robin instead of at random, so all 975 edits are covered uniformly, and every sample records the edit it belongs to.
- **Metric correction.** The check for mentions of the old fact used a plain substring search and counted, among others, matches inside subject names. It now matches on word boundaries and skips locality tests, where the old answer is often the correct one.
- **Student model.** The student was switched to Qwen2.5-7B-Instruct, the model the published baselines edit. A new dataset of 36,864 samples was synthesized with the corrected pipeline; after both filter stages 30,442 samples remain. The LoRA adapter was retrained on this dataset ($r = 64$, $\alpha = 256$, learning rate $2 \cdot 10^{-5}$, 5 epochs).

The retrained adapter was evaluated on all 975 CounterFact edits. Table 3 reports the AKEW protocol metrics in the format of Table 2, so they can be compared directly with the interim results and the published baselines.

Method	Original		Paraphrased	
	BERTScore	ROUGE-L	BERTScore	ROUGE-L
LoRA (ours, retrained)	91.26	47.28	88.38	31.97

Table 3: AKEW CounterFact results of the retrained LoRA adapter (Qwen2.5-7B-Instruct, cleaned dataset) on all 975 edits, in the format of Table 2.

Table 4 breaks the evaluation down by test type, 6,825 test cases in total: per edit one efficacy question, two paraphrases, two locality probes drawn from the neighborhood and attribute prompts, and two portability questions drawn from the generation prompts.

Test type	BERTScore	ROUGE-L	Exact match	Judge (0–5)	Success	Old fact leak
Efficacy	91.26	47.28	83.4 %	4.44	85.4 %	14.4 %
Generalization	88.38	31.97	54.3 %	3.98	72.6 %	26.0 %
Locality	85.26	19.85	22.2 %	4.84	96.0 %	–
Portability	88.51	28.26	56.0 %	3.54	62.6 %	20.7 %
Overall	87.94	29.64	49.7 %	4.17	78.3 %	15.4 %

Table 4: Evaluation of the retrained LoRA adapter (Qwen2.5-7B-Instruct, cleaned dataset) on all 975 CounterFact edits (975 efficacy tests, 1,950 tests for each other type). Success is the share of judge scores of 4 or higher; the old fact check is skipped for locality tests, where the old answer is the correct one.

The edits themselves are learned reliably: on the original questions the adapter reaches a judge score of 4.44 and answers 83.4 percent of them with the exact new target. Locality is largely preserved at 96 percent success, so the training rarely damages unrelated facts. Generalization and portability are weaker, and in about a quarter of the paraphrase answers the old fact still appears. On the AKEW protocol (Table 3) the adapter improves on the interim adapter in all four values. Since the student is now Qwen2.5-7B-Instruct, these numbers are also directly comparable to Table 2: the adapter surpasses all locate and edit baselines on BERTScore, exceeds AlphaEdit on the original ROUGE-L, and keeps its small gap between original and paraphrased questions, while AnyEdit remains ahead on the original questions.

References

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